

Ray Lui

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Education

California Polytechnic State University - San Luis Obispo

Sep 2021 – Dec 2025

Bachelor of Science in Computer Engineering, Cum Laude (GPA: 3.65)

San Luis Obispo, California

Relevant Coursework: Data Structures and Algorithms, Discrete Structures, Object-Oriented Programming, Assembly Language Programming, Systems Programming, Operating Systems, Software Engineering, Database Systems, Machine Learning for Embedded Systems, Computer Networks, Digital Design, Computer Design, Computer Architecture, Microcontrollers and Embedded Systems, Computer Security, Distributed Computing

Experience

Lawrence Livermore National Laboratory

Apr 2026 – Present

Software Engineer

Livermore, California

- Led the architectural redesign and modernization of a laboratory-wide budgeting application by migrating from Play Framework and AngularJS to modern **Angular and Spring Boot**, implementing lazy loading and refactoring business logic into backend services to reduce application load times from **over 60 seconds to under 2 seconds**.
- Championed and integrated **D3.js**-based data visualization solutions capable of rendering and interacting with **10,000+ data nodes** simultaneously, increasing application scalability from a previous limit of approximately 100 nodes.
- Leading the modernization and consolidation of two legacy laboratory funding applications by migrating Java/Play applications to **Spring Boot and Angular**, consolidating shared funding approval workflows into a common backend while maintaining separate user interfaces for research and development and investment funding.

Software Engineer Intern

Jun 2025 – Dec 2025

- Migrated a legacy internal facility and employee management application from Oracle ADF to modern **Angular and Spring Boot** architecture, optimizing backend business logic and data handling workflows to improve response performance by **67%** (from 3 seconds to under 1 second).
- Engineered scalable full-stack features using **Angular, Spring Boot, JPA entity mapping, RESTful APIs**, and optimized **Oracle SQL** queries, enabling efficient handling of user data for **10,000+ laboratory employees** with minimal latency.
- Spearheaded the architectural redesign and development of **10+ Oracle SQL views** to optimize complex data retrieval workflows, improving initial query fetch performance by **75%** for large-scale application datasets.

Conspire Commerce

Nov 2023 – Feb 2024

Software Engineer Intern

San Luis Obispo, California

- Developed and shipped **5 production-ready pages** for an e-commerce web application using **React, TypeScript, and HTML/CSS**, translating Figma designs into responsive user interfaces.
- Implemented reusable **React components** and frontend functionality within an existing application backed by MongoDB, maintaining consistency with established design and development patterns.
- Worked in a 2-week Agile sprint cycle, collaborating with developers and stakeholders to complete assigned features **3–4 days ahead of sprint deadlines**.

Projects

MediMind - AI Health Assistant | *Python, FastAPI, React, TypeScript, PostgreSQL, LangGraph* Jan 2026 – Mar 2026

- Developed a full-stack AI health assistant using **FastAPI, React, TypeScript, PostgreSQL, and LangGraph**, featuring real-time conversational support and a comprehensive health dashboard spanning **5 tracking domains**: vitals, medications, symptoms, hydration, and personalized recommendations.
- Designed and implemented an agentic AI workflow using **LangGraph** with parallel chatbot and summarization agents, leveraging **PostgreSQL checkpointing** to maintain persistent conversational state across user sessions.
- Integrated real-time AI response streaming using **Server-Sent Events (SSE)** in **FastAPI**, injecting user health context into conversational pipelines to enable personalized AI-driven interactions.

Poly Properties - Cal Poly Rental Property Platform | *TypeScript, React, Next.js, Firebase* Jan 2025 – Jun 2025

- Architected full-stack rental listing interfaces using **React, Next.js, and Firebase**, enabling rendering and filtering of **10,000+ property records**, user reviews, pricing analytics, and housing data for Cal Poly student rentals.
- Built scalable application functionality including routing, reusable React components, asynchronous API integration, and optimized **Firebase data retrieval** with pagination to maintain **sub-second page load times** across the platform.
- Contributed to a 4 person Agile development team, participating in sprint planning, iterative feature development, and efficient project execution that consistently delivered sprint objectives.

Technical Skills

Languages: Python, Java, C, HTML/CSS, JavaScript, TypeScript, SQL

Technologies/Frameworks: Git, React, Spring Boot, Angular, Firebase, PostgreSQL, Supabase, FastAPI, Node.js, REST APIs, MongoDB, Oracle SQL, LangGraph, Next.js, Linux